Solutions: Additional Exercises 3

Linear Approximation, Total Differential & Chain Rule

Exercise 1: Total differential

(a) $z = x^3y - y^3x$: $dz = (3x^2y - y^3)\,dx + (x^3 - 3xy^2)\,dy$.

(b) $z = e^x \sin(xy)$: $z_x = e^x \sin(xy) + e^x y \cos(xy) = e^x(\sin(xy) + y\cos(xy))$, $z_y = e^x \cdot x\cos(xy)$. $dz = e^x(\sin(xy) + y\cos(xy))\,dx + xe^x \cos(xy)\,dy$.

(c) $u = x^2 + y^2 + z^2$: $du = 2x\,dx + 2y\,dy + 2z\,dz$.

(d) $z = x^2y^3$: $dz = 2xy^3\,dx + 3x^2y^2\,dy$.

(e) $z = \arctan(y/x)$: $z_x = -y/(x^2 + y^2)$, $z_y = x/(x^2 + y^2)$. $dz = \frac{-y\,dx + x\,dy}{x^2+y^2}$.

(f) $u = xyz + x^2z$: $u_x = yz + 2xz$, $u_y = xz$, $u_z = xy + x^2$. $du = (yz + 2xz)\,dx + xz\,dy + (xy + x^2)\,dz$.

(g) $z = \ln\left(\frac{x+y}{x-y}\right) = \ln(x+y) - \ln(x-y)$. $z_x = \frac{1}{x+y} - \frac{1}{x-y} = \frac{-2y}{x^2-y^2}$, $\quad z_y = \frac{1}{x+y} + \frac{1}{x-y} = \frac{2x}{x^2-y^2}$. $dz = \frac{-2y\,dx + 2x\,dy}{x^2-y^2}$.

(h) $u = \sqrt{x^2 + y^2 + z^2} = r$: $du = \frac{x\,dx + y\,dy + z\,dz}{\sqrt{x^2+y^2+z^2}}$.

(i) $z = x^y = e^{y \ln x}$: $z_x = yx^{y-1}$, $z_y = x^y \ln x$. $dz = yx^{y-1}\,dx + x^y \ln x\,dy$.

(j) $u = f(x^2y,\, y^2z)$. Let $s = x^2y$, $t = y^2z$. $u_x = f_s \cdot 2xy$, $u_y = f_s \cdot x^2 + f_t \cdot 2yz$, $u_z = f_t \cdot y^2$. $du = 2xyf_s\,dx + (x^2f_s + 2yzf_t)\,dy + y^2f_t\,dz$.

Exercise 2: Approximation

(a) $f(x,y) = x^y$; base point $(a,b) = (1,4)$, $\Delta x = 0.02$, $\Delta y = 0.05$. $f(1,4) = 1$. $f_x = yx^{y-1}$: at $(1,4)$, $f_x = 4$. $f_y = x^y \ln x$: at $(1,4)$, $f_y = 0$. $1.02^{4.05} \approx 1 + 4(0.02) + 0(0.05) = 1.08$.

(b) $f(x,y) = \ln(x^2 + y^2)$; base $(0.09, 0.99) \approx (0,1)$, $\Delta x = 0.09$, $\Delta y = -0.01$. Wait, let $a = 0, b = 1$. But $f(0,1) = \ln 1 = 0$. $f_x = 2x/(x^2 + y^2)|_{(0,1)} = 0$. $f_y = 2y/(x^2 + y^2)|_{(0,1)} = 2$. $\ln(0.09^2 + 0.99^2) \approx 0 + 0 \cdot (0.09) + 2 \cdot (-0.01) = -0.02$.

(c) $f(x,y) = \arctan(y/x)$; base $(a,b) = (1,1)$, $\Delta x = 0.05$, $\Delta y = -0.02$. $f(1,1) = \pi/4$. $f_x = -y/(x^2 + y^2)|_{(1,1)} = -1/2$. $f_y = x/(x^2 + y^2)|_{(1,1)} = 1/2$. $\arctan(1.05/0.98) \approx \pi/4 + (-1/2)(0.05) + (1/2)(-0.02) = \pi/4 - 0.025 - 0.01 = \pi/4 - 0.035$.

(d) $f(x,y)=x^3y^2$; base $(1,1)$, $\Delta x=0.02$, $\Delta y=-0.03$. $f(1,1)=1$. $f_x=3x^2y^2|_{(1,1)}=3$. $f_y=2x^3y|_{(1,1)}=2$. $1.02^3\cdot 0.97^2\approx 1+3(0.02)+2(-0.03)=1+0.06-0.06=1$.

(e) $f(x,y)=\sqrt{5e^x+y^2}$; base $(0,2)$, $\Delta x=0.02$, $\Delta y=0.03$. $f(0,2)=\sqrt{5+4}=3$. $f_x=\frac{5e^x}{2\sqrt{5e^x+y^2}}|_{(0,2)}=\frac{5}{6}$. $f_y=\frac{2y}{2\sqrt{5e^x+y^2}}|_{(0,2)}=\frac{4}{6}=\frac{2}{3}$. $\sqrt{5e^{0.02}+2.03^2}\approx 3+\frac{5}{6}(0.02)+\frac{2}{3}(0.03)=3+\frac{0.1}{6}+0.02=3+0.01\overline{6}+0.02\approx 3.037$.

(f) $f(x,y)=\sqrt[4]{x^2+y^2}=(x^2+y^2)^{1/4}$; base $(4,3)$, $\Delta x=0.05$, $\Delta y=-0.07$. $f(4,3)=(16+9)^{1/4}=25^{1/4}=\sqrt{5}$. $f_x=\frac{2x}{4(x^2+y^2)^{3/4}}|_{(4,3)}=\frac{8}{4\cdot 25^{3/4}}=\frac{2}{25^{3/4}}$. $25^{3/4}=(\sqrt{5})^3=5\sqrt{5}$. $f_x=\frac{2}{5\sqrt{5}}$. Similarly $f_y=\frac{2\cdot 3}{4\cdot 5\sqrt{5}}=\frac{6}{4\cdot 5\sqrt{5}}=\frac{3}{10\sqrt{5}}$. $\sqrt[4]{4.05^2+2.93^2}\approx\sqrt{5}+\frac{2}{5\sqrt{5}}(0.05)+\frac{3}{10\sqrt{5}}(-0.07)=\sqrt{5}+\frac{0.1}{5\sqrt{5}}-\frac{0.21}{10\sqrt{5}}.=\sqrt{5}+\frac{0.2-0.21}{10\sqrt{5}}=\sqrt{5}-\frac{0.01}{10\sqrt{5}}=\sqrt{5}-\frac{0.001}{\sqrt{5}}$. $\approx 2.2361-0.000447\approx 2.2357$.

Exercise 3: Measurement error

(a) $S=ab$. $|\Delta S|\approx|b||\Delta a|+|a||\Delta b|=9(0.1)+6(0.1)=0.9+0.6=1.5\,\text{cm}^2$.

(b) $V=\pi h(R^2-r^2)$. $V_R=2\pi hR$, $V_r=-2\pi hr$, $V_h=\pi(R^2-r^2)$. $|\Delta V|\approx 2\pi(7)(5)(0.08)+2\pi(7)(4)(0.08)+\pi(25-16)(0.2).=2\pi(7)(0.08)(5+4)+\pi(9)(0.2)=2\pi(7)(0.08)(9)+0.2\pi(9).=2\pi(5.04)+1.8\pi=10.08\pi+1.8\pi=11.88\pi\approx 37.32\,\text{cm}^3$.

(c) $\alpha=\arctan(b/a)$. $\alpha_a=-b/(a^2+b^2)$, $\alpha_b=a/(a^2+b^2)$. At $a=3$, $b=4$: $a^2+b^2=25$. $\alpha_a=-4/25$, $\alpha_b=3/25$. $|\Delta\alpha|\approx\frac{4}{25}(0.01)+\frac{3}{25}(0.01)=\frac{7}{25}(0.01)=0.0028\,\text{rad}\approx 0.16°$.

(d) $V=a^2b$. $V_a=2ab$, $V_b=a^2$. $|\Delta V|\approx 2(20)(10)(0.4)+400(0.2)=160+80=240\,\text{cm}^3$. $V=400(10)=4000\,\text{cm}^3$. Relative error: $\frac{|\Delta V|}{V}\approx\frac{240}{4000}=0.06=6\%$.

Exercise 4: Chain rule

(a) $z=u^2\ln v$, $u=y/x$, $v=x^2+y^2$. $\partial z/\partial x=\partial z/\partial u\cdot\partial u/\partial x+\partial z/\partial v\cdot\partial v/\partial x=2u\ln v\cdot(-y/x^2)+u^2/v\cdot 2x.=2(y/x)\ln(x^2+y^2)\cdot(-y/x^2)+(y^2/x^2)\cdot\frac{2x}{x^2+y^2}=\frac{-2y^2\ln(x^2+y^2)}{x^3}+\frac{2y^2}{x(x^2+y^2)}$.

(b) $u=\ln(x+y+z)$, $x=\cos^2 t$, $y=2t\sin t$, $z=t$. $\frac{du}{dt}=\frac{1}{x+y+z}\left(\frac{dx}{dt}+\frac{dy}{dt}+\frac{dz}{dt}\right)$. $x'=-2\cos t\sin t=-\sin 2t$, $y'=2\sin t+2t\cos t$, $z'=1$. $u'=\frac{-\sin 2t+2\sin t+2t\cos t+1}{\cos^2 t+2t\sin t+t}$.

(c) $z=2^{x-3y}$, $x=s^2t$, $y=st^2$. $\partial z/\partial s=\ln 2\cdot 2^{x-3y}(x_s-3y_s)=\ln 2\cdot 2^{s^2t-3st^2}(2st-3t^2)$. $\partial z/\partial t=\ln 2\cdot 2^{x-3y}(x_t-3y_t)=\ln 2\cdot 2^{s^2t-3st^2}(s^2-6st)$.

(d) $w=e^{xy}+x$, $x=u+v$, $y=e^{u+v}$. $w_u=e^{xy}(yx_u+xy_u)+x_u=e^{xy}(e^{u+v}+xe^{u+v})+1=e^{xy}e^{u+v}(1+x)+1$. w_v is identical: $w_v=e^{xy}e^{u+v}(1+x)+1$.

(e) $F=f(x^2-y^2,e^{xy})$. Let $s=x^2-y^2$, $t=e^{xy}$. $F_x=f_s\cdot 2x+f_t\cdot ye^{xy}$. $F_y=f_s\cdot(-2y)+f_t\cdot xe^{xy}$.

(f) $u=f(x,y)$, $x=r\cos\theta$, $y=r\sin\theta$. $u_r=f_x\cos\theta+f_y\sin\theta$, $u_\theta=-f_xr\sin\theta+f_yr\cos\theta$. Then $u_r^2+\frac{1}{r^2}u_\theta^2=f_x^2\cos^2\theta+2f_xf_y\cos\theta\sin\theta+f_y^2\sin^2\theta+f_x^2\sin^2\theta-2f_xf_y\cos\theta\sin\theta+f_y^2\cos^2\theta=f_x^2+f_y^2$. So $\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2=\left(\frac{\partial u}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial u}{\partial\theta}\right)^2$.

(g) $z = f(x + ay)$. Let $w = x + ay$. Then $z_x = f'(w) \cdot 1 = f'(w)$ and $z_y = f'(w) \cdot a = af'(w) = az_x$. ✓

(h) $z = y\varphi(x^2 - y^2)$. Let $t = x^2 - y^2$. $z_x = y\varphi'(t) \cdot 2x$. $z_y = \varphi(t) + y\varphi'(t)(-2y) = \varphi(t) - 2y^2\varphi'(t)$. $\frac{z_x}{x} = 2y\varphi'(t)$. $\frac{z_y}{y} = \frac{\varphi(t)}{y} - 2y\varphi'(t)$. $\frac{z_x}{x} + \frac{z_y}{y} = \frac{\varphi(t)}{y} = \frac{z}{y \cdot y} = \frac{z}{y^2}$ (since $z = y\varphi(t)$, so $\varphi(t) = z/y$). ✓

Exercise 5: Implicit differentiation

(a) $F = x^2 + y^2 + z^2 - 1$. $F_x = 2x$, $F_y = 2y$, $F_z = 2z$. $z_x = -F_x/F_z = -x/z$, $z_y = -y/z$.

(b) $F = x^2 + y^2 - z$. $F_x = 2x$, $F_y = 2y$, $F_z = -1$. $z_x = 2x$, $z_y = 2y$.

(c) $F = e^{\sin z} - z^2xy$. $F_x = -z^2y$, $F_y = -z^2x$, $F_z = e^{\sin z}\cos z - 2xyz$. $z_x = -(-z^2y)/(e^{\sin z}\cos z - 2xyz) = \frac{z^2y}{e^{\sin z}\cos z - 2xyz}$. $z_y = \frac{z^2x}{e^{\sin z}\cos z - 2xyz}$.

(d) $F = xy + yz + xz - 1$. $F_x = y + z$, $F_y = x + z$, $F_z = y + x$. $z_x = -(y + z)/(x + y)$, $z_y = -(x + z)/(x + y)$.

(e) $\Phi(cx - az, cy - bz) = 0$. Let $u = cx - az$, $v = cy - bz$. Differentiate w.r.t. x: $\Phi_u(c - az_x) + \Phi_v(-bz_x) = 0 \Rightarrow z_x(-a\Phi_u - b\Phi_v) = -c\Phi_u$... Hmm, rather: $\Phi_u(c - az_x) + \Phi_v(0 - bz_x) = 0 \Rightarrow z_x(a\Phi_u + b\Phi_v) = c\Phi_u$. Differentiate w.r.t. y: $\Phi_u(-az_y) + \Phi_v(c - bz_y) = 0 \Rightarrow z_y(a\Phi_u + b\Phi_v) = c\Phi_v$. $az_x + bz_y = \frac{c(a\Phi_u + b\Phi_v)}{a\Phi_u + b\Phi_v} = c$. ✓

(f) $F = z\ln(x+z) - xy/z$. $F_x = z \cdot \frac{1}{x+z} - y/z$, $F_y = -x/z$, $F_z = \ln(x+z) + z \cdot \frac{1}{x+z} + xy/z^2$.

$$z_x = -F_x/F_z = \frac{y/z - z/(x+z)}{\ln(x+z) + 1/(x+z) \cdot z + xy/z^2} = \frac{y(x+z) - z^2}{z(x+z)\left[\ln(x+z) + z/(x+z) + xy/z^2\right]}.$$

$$z_y = -F_y/F_z = \frac{x/z}{\ln(x+z) + z/(x+z) + xy/z^2}.$$

(g) $F(x - z, y - z) = 0$. Let $s = x - z$, $t = y - z$. $\frac{\partial}{\partial x}$: $F_s(1 - z_x) + F_t(0 - z_x) = 0 \Rightarrow F_s - z_x(F_s + F_t) = 0 \Rightarrow z_x = \frac{F_s}{F_s + F_t}$. $\frac{\partial}{\partial y}$: $F_s(-z_y) + F_t(1 - z_y) = 0 \Rightarrow F_t - z_y(F_s + F_t) = 0 \Rightarrow z_y = \frac{F_t}{F_s + F_t}$. $z_x + z_y = \frac{F_s + F_t}{F_s + F_t} = 1$. ✓